SEA.AI

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1. Welcome

Welcome to the SEA.AI Watchkeeper Series. This system enhances maritime safety by providing advanced object detection and situational awareness through artificial intelligence. Designed for motorboats, commercial vessels, and yachts, Watchkeeper continuously monitors your surroundings and alerts operators to potential hazards.



Thank you for choosing SEA.AI.

Our system integrates cutting-edge machine vision, deep learning algorithms, and high-performance thermal and optical cameras to deliver reliable detection—day or night. It detects objects often missed by Radar or AIS, such as floating containers, buoys, kayaks, inflatables, and even people in the water.

SEA.AI system's artificial intelligence is based on the latest Machine Vision technology, best-in-class deep learning capabilities, and a constantly evolving, proprietary database of millions of annotated marine objects. Our software is regularly updated to improve functionality and reliability. Keep your device up to date in order to secure the best performance of the system.

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The SEA.AI System is intended to provide additional assistance to the User and does not exempt the User from strict compliance with the usual rules and practices of navigation safety and international maritime regulations. Users must be vigilant at all times and ensure that they have the required knowledge in terms of navigation. They must not rely on the use of the SEA.AI System to make navigation choices. Ensure the system is installed by certified professionals. Do not attempt to open or modify internal components.

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2. Product Overview and Specifications

Watchkeeper – Key Specifications

Watchkeeper provides **high-quality colour vision for day and low-light conditions** .

Technical specifications:

Camera sensor	Camera resolution	FoV	Detection Range (Buoy/RIB/Yacht)	Size	Weight	Power	Other features
4K Low Light with HDR	3840x2160	130°x65°	65m/350m/900m	110x163x86mm	935g	12-36V DC 26.1W	IP67 waterproof, IEC 60945 compliant, Ethernet/GPS/LTE

Watchkeeper's Thermal Options

Depending on the selected configuration, Watchkeeper will have different thermal camera(s) and can achieve different detection ranges and performances.

For Motorboats

All specifications below refer to thermal vision for powerboats.

Thermal Resolution	IR Cameras	FoV	Detection Range (Buoy/RIB/Yacht)	Weight	Power
320x256	1	24° x 19°	110m/500m/1.300m	951g	27.1W
640x512	1	24° x 19°	220m/960m/2.500m	975g	27.7W
1280x512	2	48° x 19°	220m/960m/2.500m	1023g	29.3W

All other specifications (color sensor, size, connectivity, waterproofing) remain the same as the ones previously indicated.

For Sailing boats

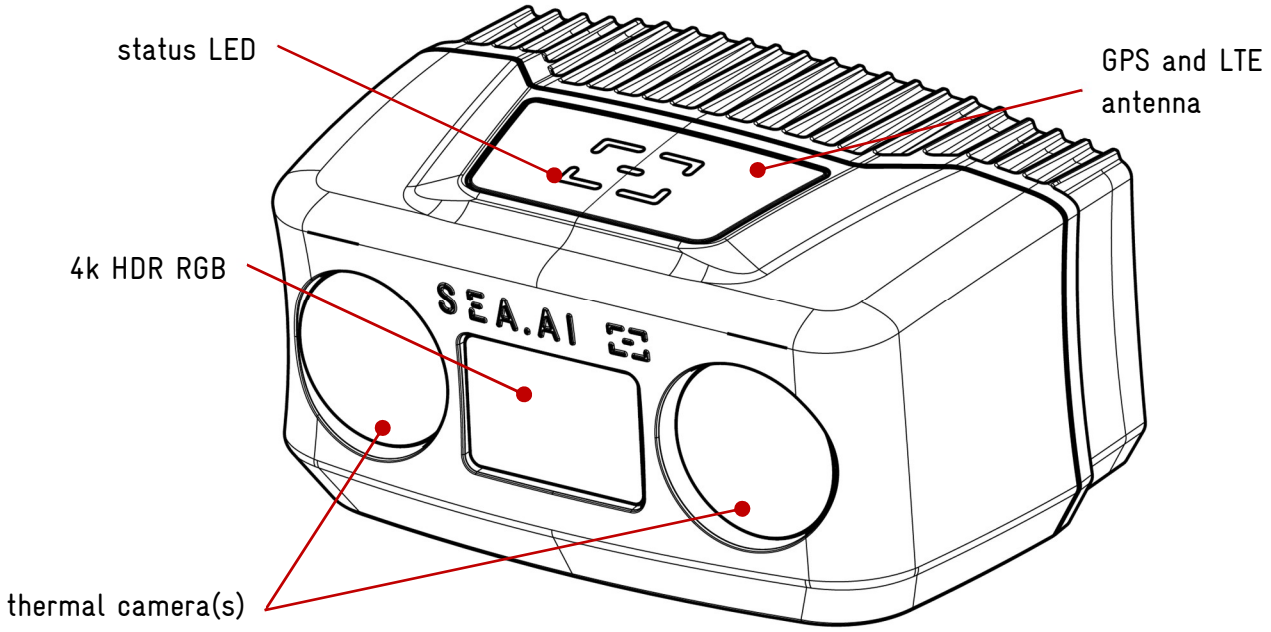
All specifications below refer to thermal vision for sailing yachts.

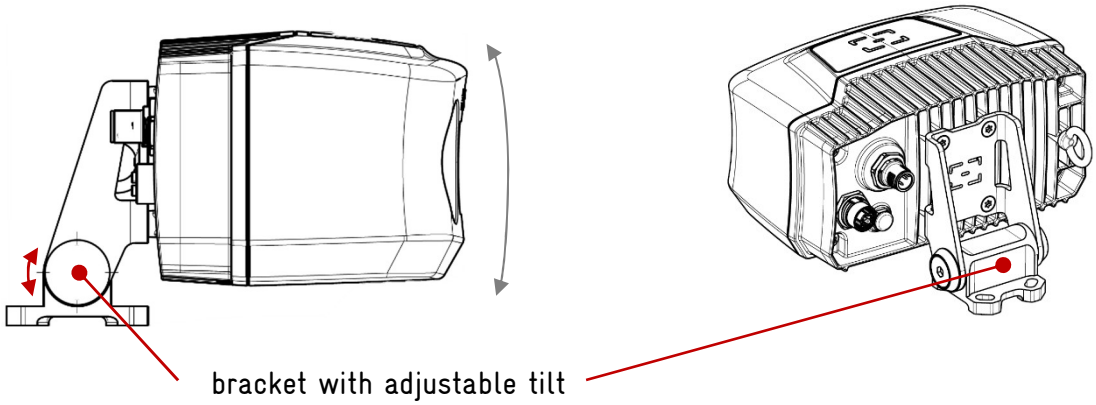
Thermal Resolution	IR Cameras	FoV	Detection Range (Buoy/RIB/Yacht)	Weight	Power
320x256	1	34° x 25°	80m/350m/900m	951g	27.1W
640x512	1	50° x 40°	110m/460m/1.200m	975g	27.7W
1024x640	2	50° x 34°	170m/750m/2.000m	1023g	29.3W

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The SEA.AI Watchkeeper Series includes:

- Vision Unit with integrated AI edge computing processor
- 4k HDR low-light optical color camera and Long-wave IR Thermal camera(s) (configuration depends on Watchkeeper model- see table above)
- GPS and LTE connectivity module
- Power and Ethernet connector





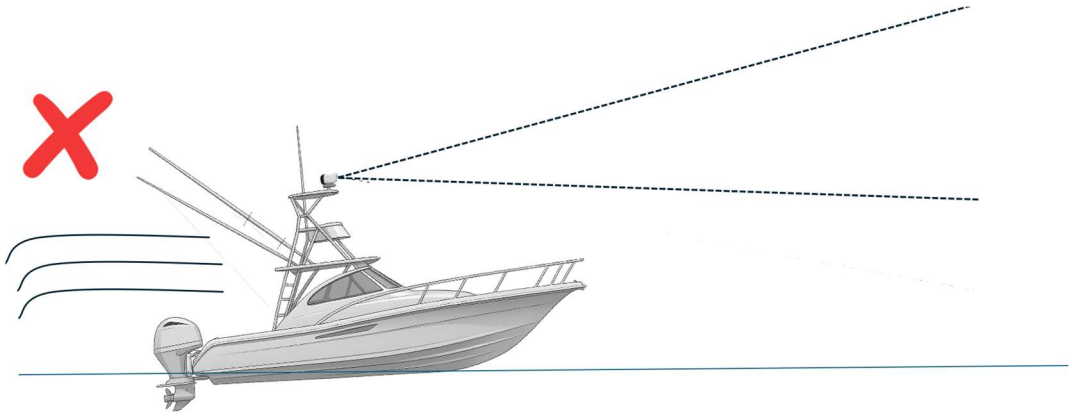
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3. Installation

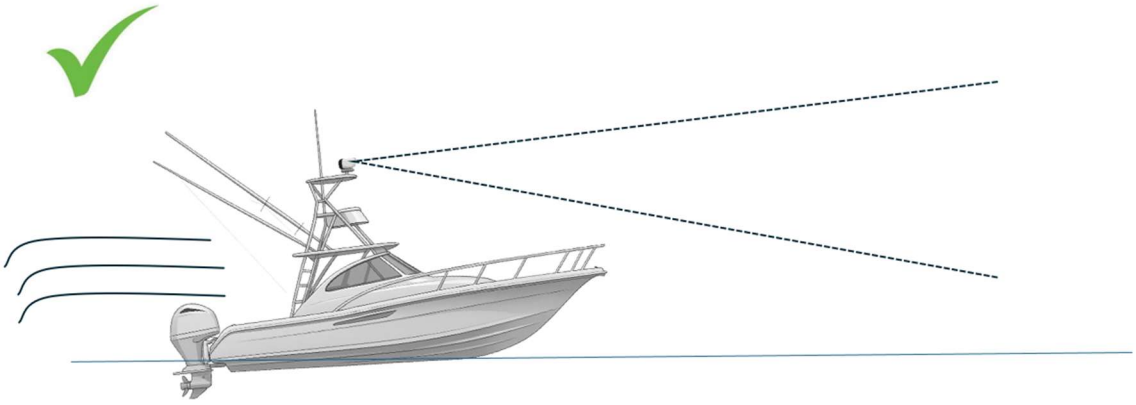
Note: Installation must be completed by a certified SEA.AI installer. Please check the installation guide in the SEA.AI document library. Accessible under www.sea.ai/document-library/ in the section Watchkeeper.

Positioning Tips:

- Mount the camera in a high location with a clear, unobstructed field of view.
- Adjust the tilt angle by rotating the camera to compensate for boat trim at cruising speed.
- Use the thermal camera as the primary reference during positioning.



Watchkeeper looking too high



Watchkeeper looking straight ahead. Horizon must be visible at cruising speed.
(It is recommended to have 20% to 30% of the thermal image above the horizon)

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4. Operating Instructions

Product registration and activation:

To ensure your new SEA.AI Watchkeeper delivers optimal performance and stays up to date with the latest advancements in Artificial Intelligence, periodic updates are essential.
To receive Software Updates and your product key to activate your SEA.AI Watchkeeper you need to register your product.

- Register your product easily by logging into your SEA.AI Connect account at: https://connect.sea.ai/account/login
- If you do not have an account yet, you can simply create one at: https://connect.sea.ai/account/register

Powering On:

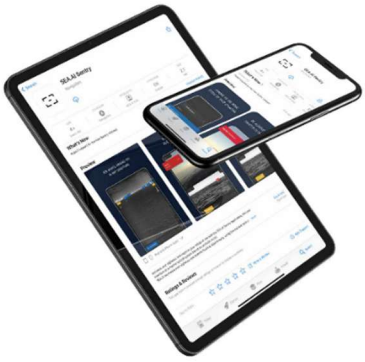
- Ensure all components are connected (see connection diagram in Installation guide).
- Power on the system via the main control panel.

Connecting External Devices:

- How to connect your mobile devices (phone, tablet)?
    - Download the latest version of the SEA.AI App.

App Store:

Google Play Store:



    - Connect to the SEA.AI app using your MFD WIFI or your boat WIFI via router to access the SEA.AI Watchkeeper.
    - When connected to the Wi-Fi, the SEA.AI App will automatically find your new Watchkeeper Camera. Please make sure your device stays connected to the Wi-Fi.

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Connecting Onboard PC:

- Download the Sea.AI Watchkeeper Windows/Mac application from Sea.AI Document library
- Open the application. Your PC will find automatically the Watchkeeper if it's connected to the same network.

Connecting Onboard MFD:

- The computing power of your MFD might be limiting our ability to display the best version of our application, therefore we adapt to give you the best out of it. Depending on your MFD brand and model, our connection protocol and user interface will differ to ensure the best performance.
- Please refer to our MFD compatibility page on our website: https://www.sea.ai/product-watchkeeper/mfd-compatibility/
- To start the SEA.AI App on your MFD – go to 3rd party Apps and tap on the SEA.AI Icon. The Icon will appear automatically once SEA.AI is connected and turned on. Start-up of the system and discovery in the MFD network can take several minutes.
- To directly access the RTSP Video with your MFD please check your MFD manual (Use the guide to setup RTSP streams on Furuno, B&G or Simrad.)
- RTSP streams:

Watchkeeper Lite:

Camera	Link	Resolution	FoV (H°xV°)
Color	rtsp://{IP}:8555/rgb1	3600x1080	130°x65°

Watchkeeper 320:

Camera	Link	Resolution	FoV (H°xV°)
Thermal	rtsp://{IP}:8555/t1	320x256	24°x19°
Color	rtsp://{IP}:8555/rgb1	3600x1080	130°x65°

Watchkeeper 640:

Camera	Link	Resolution	FoV (H°xV°)
Thermal	rtsp://{IP}:8555/t1	640x512	24°x19°
Color	rtsp://{IP}:8555/rgb1	3600x1080	130°x65°

Watchkeeper 1280:

Camera	Link	Resolution	FoV (H°xV°)
Thermal	rtsp://{IP}:8555/t1	1280x512	48°x19°
Color	rtsp://{IP}:8555/rgb1	3600x1080	130°x65°

Color camera RTSP stream example: rtsp://192.168.1.191:8555/rgb1

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Using the Interface:

- Views: Thermal (only for Watchkeeper 320, 640 and 1280) or Color.
    - To access the thermal view, select
    - The thermal view shows the live images from the thermal camera with the following field of view (Horizontal x Vertical):
        - Watchkeeper 320: 24°x19°
        - Watchkeeper 640: 24°x19°
        - Watchkeeper 1280: 48°x19°
    - The thermal view works day and night and is particularly adequate for night or low visibility conditions.
    - To access the color view, select
    - The Color view shows the live images from the low-light camera with a field of view of 130°x65°.
    - It works in daylight but also in low-light conditions like dawn or dusk.

- Each view will display bounding boxes around the detections.



- Navigation modes:
    - 

        - Suitable to use when entering and leaving harbours, marinas or crowded bays. The different views will show and highlight the objects detected, but all the alarms are off. Be aware of your surroundings and switch to Coastal Mode as soon as you leave the crowded space.

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- 

    - Suitable for coastal passages or navigation through areas of high traffic. This mode focuses on close proximity awareness for nearshore navigation. Acoustic alarms will be triggered only by dangerous targets.
- 

    - Suitable for offshore passages. Alarm sensitivity will be higher than for coastal mode.
- Zoom and Pan:
    - In both color and thermal view, you are able to pass from the full field of view to a narrow-focused view by selecting 
    - You can then pan left, right and come back to the middle by
        - Clicking on the arrows 
        - Navigating the cursor on the horizon bar 
    - You may come back to the wide vision by selecting
- Settings:
    - System tab will show you:
        - The current software version installed on your Watchkeeper
        - The storage space available on your unit. (For AI feedback recording).
        - Will allow you to upload the latest version of software onto your Watchkeeper. See how to update your system in the 5. Update section.

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- The Modes tab explains the differences between the 3 Navigation modes.

- The Personalization tab will allow you to:
  - Choose the interface theme on your display: Light for daylight, Dark for low-light conditions, and Low contrast for night navigation to preserve your night vision.
  - Choose the display scale to have different icon sizes and adapt best to your display size.
  - Choose the thermal polarity in your thermal view (White is Hot: Normal or Black is Hot: Inverted)
  - Enable or disable the voice alarm that will alert when an object is detected, based on its threat level and direction.
  - Enable or disable the NMEA 2000 buzzer to play sound alarms on external Buzzers.
  - Choose the distance unit used to display. You can choose between meters and feet. Over 1 nautical mile the distance will be shown in Nautical miles.

- The Setup tab will allow you to:
  - Select and configure the NMEA Gateway model to be used.
  - Enter the mounting height of your Watchkeeper and the distance to the bow of your boat.

    For heights higher than 6m, the system will display a distance estimation below the detection icon when an object is detected. The accuracy of the calculation depends a lot on the accuracy of the height, so make sure to be as precise as possible.
  - Give access to support from your dealer/installer.

- The Feedback tab will give you access to
  - Our feedback email so you can send us the recording performed using the AI feedback tool 
  - AI Feedback procedure:

    - Record raw images by clicking on the AI feedback tool .
    - Select the type of feedback that you want to share with Sea.AI.
    - The recording is on, you can stop it when you want, and it will be saved

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automatically into the system's memory.
- When you can, connect with a computer, download the file and contact us via email at feedback@sea.ai

- The help tab will give you access to our service support email address: service@sea.ai.com
  - You can enable or disable temporary remote access to our service team for system diagnostics and repair. SEA.AI cannot access your device or fetch data from your device without your consent.

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5. Update your Watchkeeper

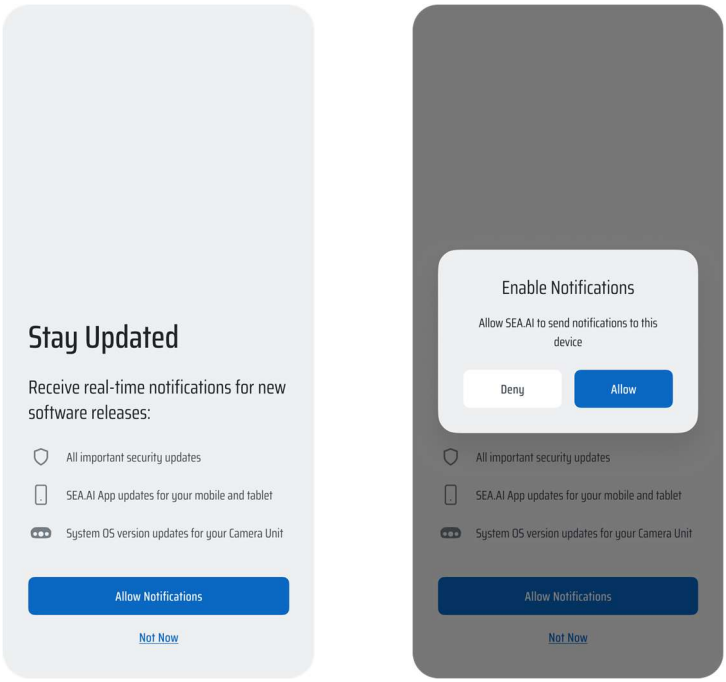
Update with your computer:

- Use your computer to download the latest Sea.AI package for your Watchkeeper on connect.sea.ai https://connect.sea.ai/account/login
- Once the package is downloaded on your computer, access your Watchkeeper app and upload it into the settings/system update section.
- If you have any problem with this procedure, please contact your local dealer/installer.

Update with your smartphone:

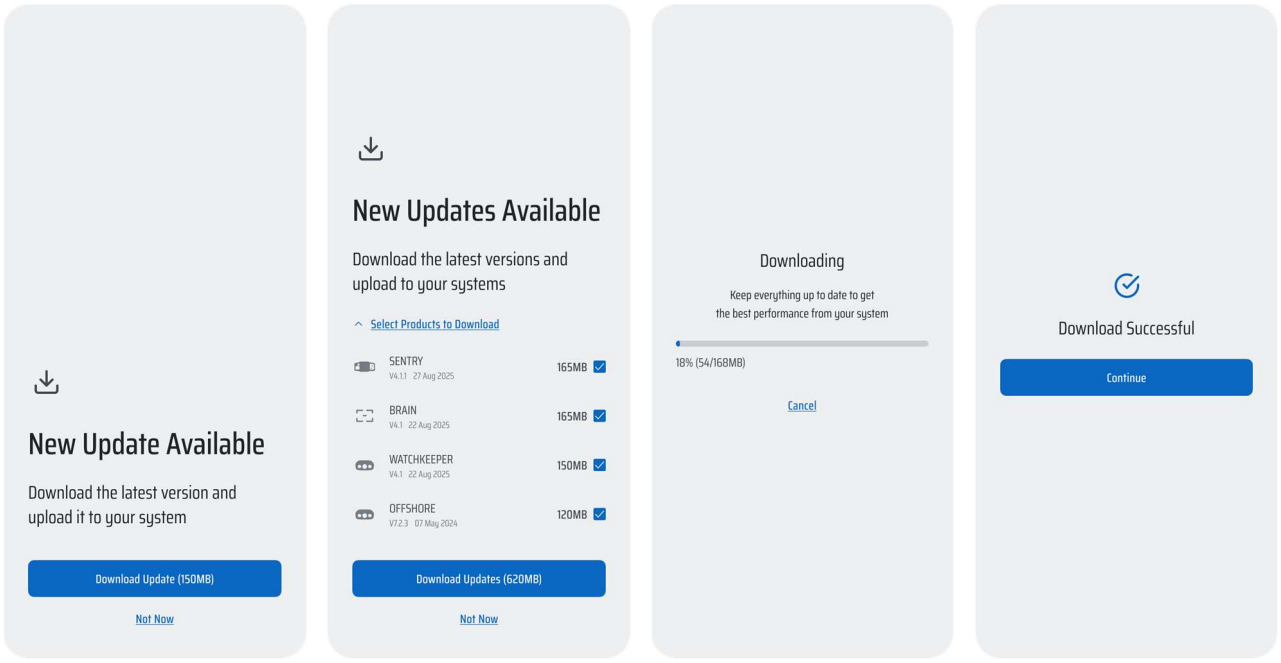
- You can also update from your smartphone. Make sure your SEA.AI application on your smartphone (Apple App Store or Google Play Store) is up to date.

  At your first connection, make sure you enable notifications for new software releases:



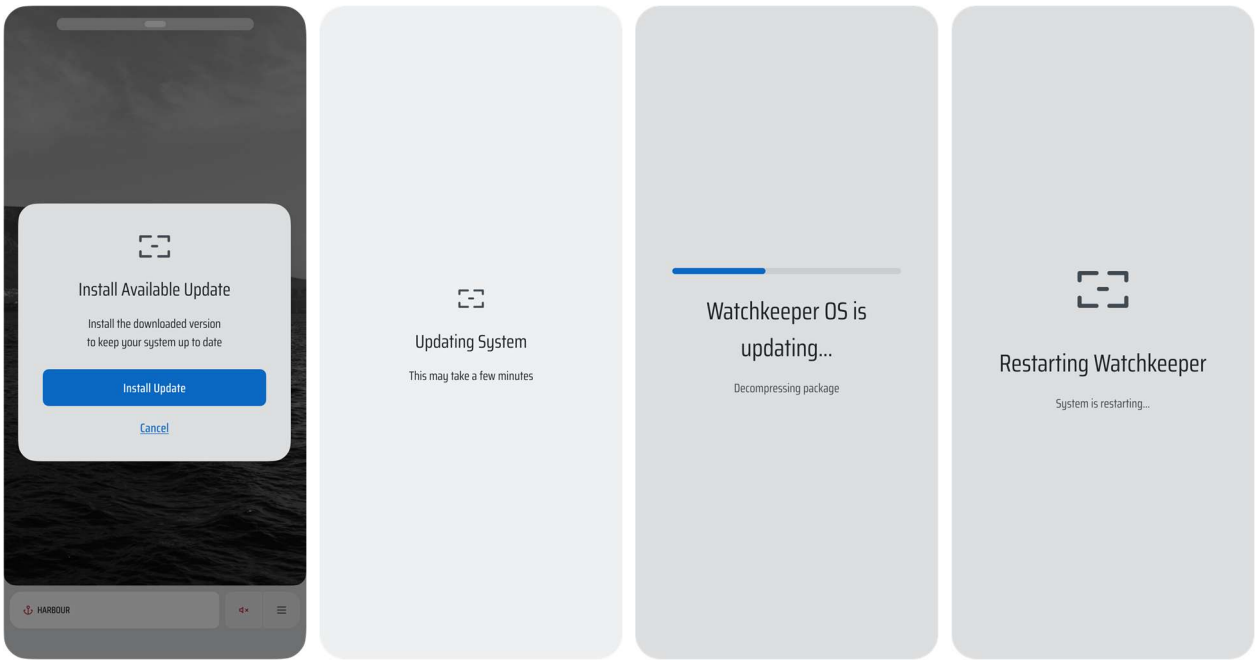
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When a new update is available you will receive a notification. Follow these steps to update your watchkeeper:



- Select download update
- If you have several systems, select the system(s) you want to update.
- Download the package.

Then apply the update to Watchkeeper:



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6. Alarms

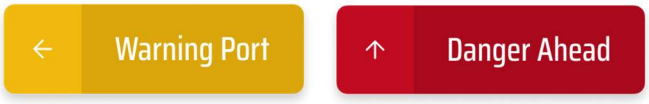
- **AI Detection:** The object classification includes 6 different classes:

Ship	
Motorboat	
Sailing boat	
Buoy	
Person in the water	
Unknown Hazard	

How the AI decides what to show

Our AI looks at every object it detects and calculates how confident it is about the object's type. If the confidence is high enough (the required level varies depending on the type of object), the system will display that class. If the confidence doesn't reach the required level, the object will simply appear as **Unknown Hazard**, so you're still aware that something is there, even if it can't be clearly identified.

- **Smart Alarms:** Our smart alarm prioritizes risks and provides clear warnings on the potential hazards you may face. It is based on the distance and bearing of the detection. A frontal detection will have a higher level of threat than a side one.
    - Alarm tags:
        - Threat levels: Object (Blue), Warning (Yellow), Danger (Red)
        - Threat Headings: Ahead, Starboard, Port

Example: 

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- Coastal alarms:

Alarm	Object	Warning	Danger
Visual			
Sound	None	2 Bips every 10s	3 Bips every 3s
Voice	None	None	"Danger (bearing)"

- Offshore alarms:

Alarm	Object	Warning	Danger
Visual			
Sound	1 Bip every 10s	2 Bips every 3s	3 Bips in loop
Voice	None	"Warning (bearing)"	"Danger (bearing)"

- Your alarm can be muted or activated using: 

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7. Maintenance and Troubleshooting

Regular Maintenance:

- Clean camera lenses weekly using approved solutions.
- Check cables integrity monthly.
- Update firmware via USB or network connection quarterly.

Troubleshooting Tips:

- *No Video Feed:* Check camera power and cable connections and reboot your system.
- *False Alarms:* Adjust detection sensitivity by changing navigation mode.
- *System Lag:* Reboot and verify system resources.
- If you encounter many false detections, please use the AI feedback and help us get better.

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8. Warranty and Support

SEA.AI provides a 2-year limited warranty covering manufacturing defects (See our Terms and conditions at www.sea.ai/termsconditions).

Activate your product and warranty by registering at: https://connect.sea.ai/account/register

For support:

- You can have access to our Frequently Asked Questions through this address: https://sea.ai/faq
- Contact your dealer or installer for any support.
- Email service@sea.ai for direct support from our Professional Services team.

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